

MOTION DETECTOR SECURITY ALARM

Using Digispark Development Board

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This circuit produces an alarm whenever someone comes near the spotting range of its passive infrared (PIR) motion sensor. It can be used as a security system to guard your home and other property, if utilised properly. The author's prototype is shown in Fig. 1.

Circuit and Working

Circuit diagram of the motion detector security alarm is shown in Fig. 2. It is built around a Digispark board, PIR sensor, 9V battery (BATT.1), resistors (R1 through R3), capacitors (C1 through C3) and a few other components.

Despite its small size, Digispark board is packed with many capabilities. Apart from having fewer I/O pins, it can do most of what the microcontroller (MCU) in Arduino board can do. It can also be programmed using Arduino IDE software.

Building the circuit is easy. You need two essential components: a PIR motion sensor module (HC-SR501) and a Digispark (Attiny85) USB development board. As both these are made to operate on batteries, you can use a standard 9V 6F22 battery, or any other 9V rechargeable battery, as the power source. Switch S1 is the master power on/off switch. Both the PIR sensor and Digispark have in-built voltage regulators and, hence, input DC supply (9V) is pre-regulated by their internal circuitry to develop required working voltages.

Output (OUT) from the PIR sensor is connected to input pin P0 of Digispark through a BC547 level inverter (T1). No pull-up resistor is required here because internal pull-up of Attiny85 is enabled. Output pin P3 of Digispark drives the LED (LED1), while pin P2 controls the switching of the piezo-buzzer (PZ1)



Fig. 1: Author's prototype

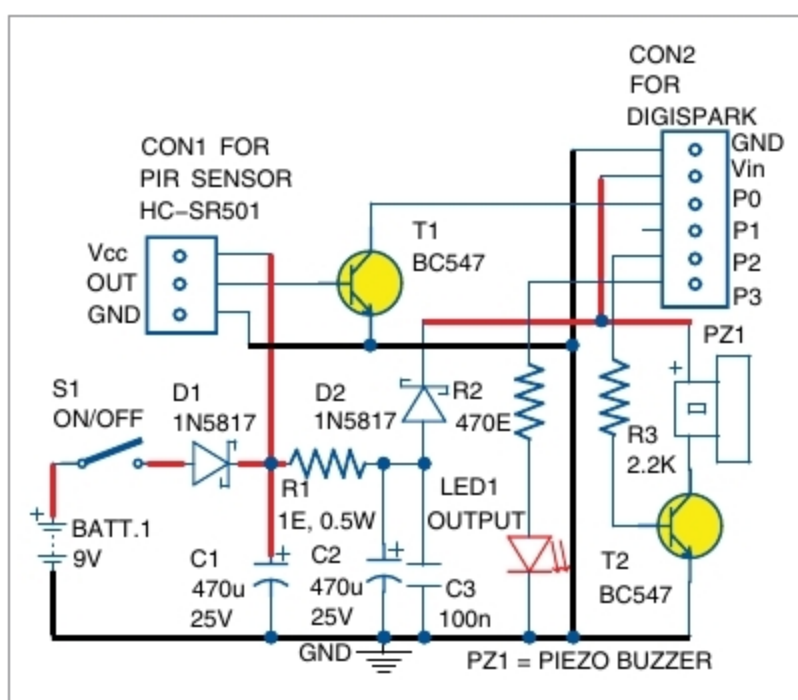


Fig. 2: Circuit diagram of motion detector security alarm

PARTS LIST

Semiconductors:

- D1-D2 - 1N5817 Schottky diode
- T1-T2 - BC547 npn transistor
- LED1 - 5mm LED
- Digispark - Digispark development board

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 1-ohm, 0.5-watt
- R2 - 470-ohm
- R3 - 2.2-kilo-ohm

Capacitors:

- C1-C2 - 470 μ F, 25V electrolytic
- C3 - 100nF ceramic disk

Miscellaneous:

- CON1 - 3-pin connector for PIR sensor
- CON2 - 6-pin connector for Digispark
- BATT.1 - 9V battery
- S1 - On/off switch
- PZ1 - Piezo-buzzer
- Jumper wires

through another transistor (T2).

Digispark board

For details on Digispark, refer Getting Started example in the documentation available at <http://digistump.com/wiki/start>. However, make a note of some significant details given below.

1. Digispark requires Arduino IDE 1.6.5 or later version.

2. Before plugging Digispark board into the USB port of your PC, or before running Arduino, download and install the requisite DigiSpark driver from <http://sourceforge.net/projects/digistump/files/Digispark-Arduino-Win32-1.0.4-May19.zip/download>

3. Do not plug in Digispark into the USB port before invoking the

upload instructions from Arduino IDE. To do that, hit Upload. The status box at the bottom will ask you to plug in Digispark. At this point, plug it in. If there is any problem, unplug it and plug it back in.

4. If you unplug Digispark and plug it back in, or attach it to another power source, there will be a delay of five seconds before your program (digi_motion.ino) will run.

5. After successfully uploading the program to Digispark, plug it out from the USB port.

Construction and testing

An actual-size PCB for the motion detector security alarm is shown in